Practical 10: Recursion and Backtracking (CO2)

Code:

```
def factorial(n):
    if n == 0:
        return 1
    return n * factorial(n - 1)

def solve_nqueens(n):
    solutions = []
    board = [-1] * n

    def is_safe(row, col):
        for r in range(row):
            if board[r] == col or abs(board[r] - col) == abs(r - row):
                return False
        return True

    def backtrack(row):
        if row == n:
            solutions.append(board.copy())
            return
        for col in range(n):
            if is_safe(row, col):
                board[row] = col
                backtrack(row + 1)
                board[row] = -1

    backtrack(0)
    return solutions

print("Factorial of 5:", factorial(5))

solutions = solve_nqueens(4)
print(f"Number of solutions for 4-Queens: {len(solutions)}")
print("First solution (col positions per row):", solutions[0])
```

Output:

```
Factorial of 5: 120
Number of solutions for 4-Queens: 2
First solution (col positions per row): [1, 3, 0, 2]
```